

CSCI 7250 – Advanced Reverse Engineering Research Project

For your Reverse Engineering Research Paper/Project, research and implement an applied topic in RE.

- Reversing and hacking an application or OS (Lin/Win/Android/iOS/ARM/console or an obscure system)
- Game cheats (your favorite first-person shooter, even just hacking Minesweeper, Plants v Zombies, etc.)
- Mobile/Web/Network RE (ESCALATE or other mobile/Web/Net/etc. reverse engineering challenges)
- Car Hacking (reversing CAN packets on a real car)
- Reversing/hacking smart home and other IoT devices (check out exploitee.rs)
- SCADA/ICS industrial/operational devices (Hack-a-Sat/Hack-the-Port/PLCs, HMIs, MODBUS, etc.)
- Firmware reversing/hacking (check out FirmwareSlap on github)
- Advanced CTF problems (Reverse Engineering and/or Binary Exploit Development)
- Any other topic of interest to you that fits under reverse engineering (see me if you have any questions)

Also, see popular conference web sites and proceedings (BlackHat, DEFCON, HackerHalted, ACM proceedings, etc.) for general ideas to research/implement/develop further.

1. Research Poster or Conference Paper Writeup (50%): Produce a short-form paper in the ACM format given below, 1.0-1.5 pages in length (using the ACM publication format - this will usually be 2-3 pages of regular typewritten text), or a research poster in the ACM PPT format given (36"x48") writing up your project. NOTE: 5% extra credit for submitting your abstract/paper/poster to a conference (including ARC, GSRC, ACMSE, etc.) and 5% more for getting accepted and presenting it there.

- Title and Abstract

2. Implementation Component (50%): Perform reverse engineering and demonstrate the results of your modified code live in class (i.e. hack the game/app/device live, or show a short video of the hack pre-recorded at home). You'll present a brief 5-slide PowerPoint (or Presi/Keynote/Google Slides/etc.) presentation summarizing your work, plus references as appropriate. For full credit, you must demonstrate **evidence** of the reverse engineering tasks you performed personally (disassembly/decompilation/debugging/patching/etc.)

Be ready to present your work on Tue, Apr 22 through the final exam night on May 6 – we'll run 8-10 demos/presentations per class – here's a signup sheet for your topics, randomly assigned – you may swap days with another person IF you have their permission:

https://docs.google.com/spreadsheets/d/16MmCygZiXJd7w6eS_WxltLC6AoN4tO9jQD9VNt69dgA/edit?usp=sharing